

Set 6: Product Sales Analysis

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

Program:

```
# Load libraries
library(ggplot2)
library(tidyr)

# Create dataset
sales <- data.frame(
  Product_ID = c(1,2,3),
  Product_Name = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
  March = c(2400,1600,1100)
)

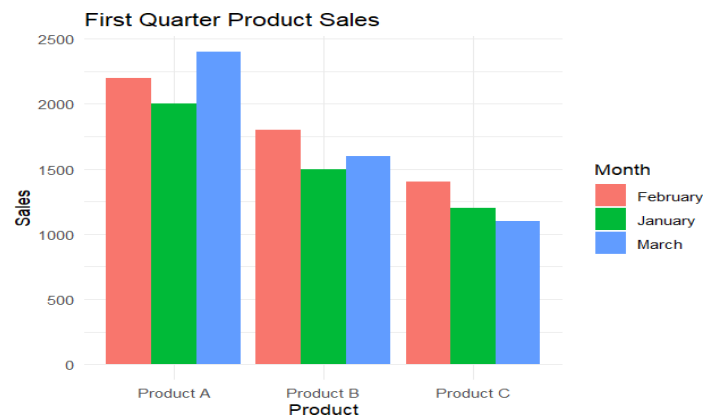
# Convert to long format
sales_long <- pivot_longer(
  sales,
  cols = c(January, February, March),
  names_to = "Month",
  values_to = "Sales"
)

# Grouped Bar Chart
ggplot(sales_long,
  aes(x = Product_Name,
    y = Sales,
    fill = Month)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(
    title = "First Quarter Product Sales",
    x = "Product",
    y = "Sales"
```

```
) +
```

```
theme_minimal()
```

Output:



2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.

Program:

```
# Load libraries
```

```
library(ggplot2)
```

```
library(tidyr)
```

```
# Create dataset
```

```
sales <- data.frame(  
  Product_Name = c("Product A", "Product B", "Product C"),  
  January = c(2000, 1500, 1200),  
  February = c(2200, 1800, 1400),  
  March = c(2400, 1600, 1100)  
)
```

```
# Convert to long format
```

```
sales_long <- pivot_longer(  
  sales,  
  cols = c(January, February, March),  
  names_to = "Month",  
  values_to = "Sales"  
)
```

```
# Order months
```

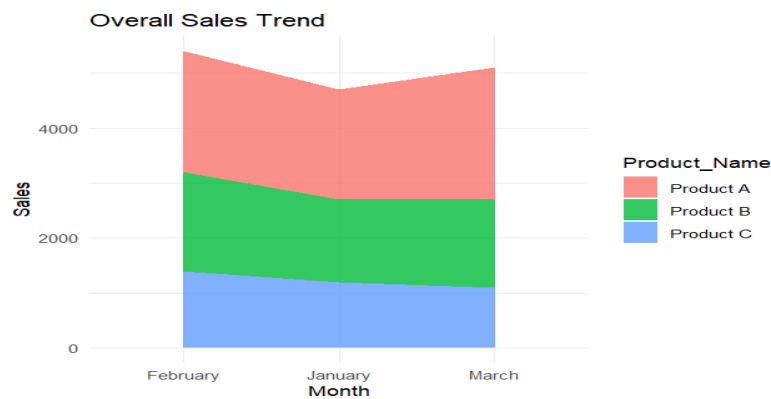
```
sales_long$Month <- factor(  
  levels = c("March", "February", "January")  
)
```

```

sales_long$Month,
levels = c("January", "February", "March")
)
# Stacked Area Chart
ggplot(sales_long,
aes(x = Month,
y = Sales,
fill = Product_Name,
group = Product_Name)) +
geom_area(alpha = 0.8) +
labs(
title = "Overall Sales Trend",
x = "Month",
y = "Sales"
) +
theme_minimal()

```

Output:



3. Build a table to show the monthly sales data for each product. Label the table elements.

Program:

```

# Create dataset
sales <- data.frame(
  Product_ID = c(1,2,3),
  Product_Name = c("Product A", "Product B", "Product C"),
  January_Sales = c(2000, 1500, 1200),
  February_Sales = c(2200, 1800, 1400),

```

```
March_Sales = c(2400,1600,1100)
)
```

```
# Display table
```

```
print(sales)
```

Output:

```
Product_Name January February March
```

```
1 Product A  2000   2200  2400
```

```
2 Product B  1500   1800  1600
```

```
3 Product C  1200   1400  1100
```

Set 7: Customer Demographics Analysis

1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.

Program:

```
# Load library
```

```
library(ggplot2)
```

```
# Create dataset
```

```
customers <- data.frame(
```

```
  Customer_ID = c(1,2,3),
```

```
  Age = c(28,35,42),
```

```
  Gender = c("Female","Male","Female"),
```

```
  Income = c(50000,60000,75000)
```

```
)
```

```
# Bar Chart
```

```
ggplot(customers,
```

```
  aes(x = factor(Customer_ID),
```

```
    y = Age,
```

```
    fill = factor(Customer_ID))) +
```

```
geom_bar(stat = "identity") +
```

```
labs(
```

```
  title = "Customer Age Distribution",
```

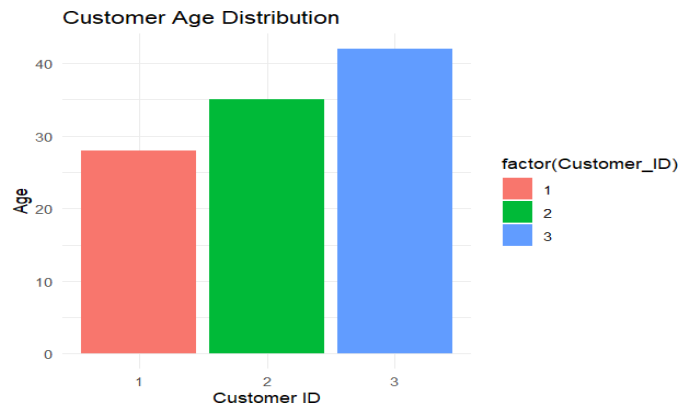
```
  x = "Customer ID",
```

```
  y = "Age"
```

```
) +
```

```
theme_minimal()
```

Output:



2. Generate a pie chart to represent the distribution of customers by gender.

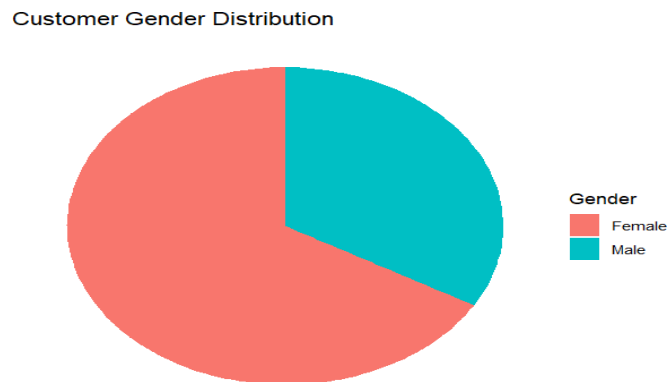
Program:

```
# Load libraries
library(ggplot2)

# Create dataset
customers <- data.frame(
  Customer_ID = c(1,2,3),
  Gender = c("Female","Male","Female")
)

# Count genders
gender_count <- as.data.frame(table(customers$Gender))
colnames(gender_count) <- c("Gender","Count")

# Pie Chart
ggplot(gender_count,
  aes(x = "",
    y = Count,
    fill = Gender)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y") +
  labs(title = "Customer Gender Distribution") +
  theme_void()
```

Output:

3. Build a table to show the demographic information for each customer. Label the table elements.

Program:

```
# Create dataset
customers <- data.frame(
  Customer_ID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)
# Display table
print(customers)
```

Output:

```
Customer_ID Gender
1          1 Female
2          2  Male
3          3 Female
```

Set 8: Employee Performance Analysis

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

Program:

```
# Load library
library(ggplot2)
# Create dataset
```

```

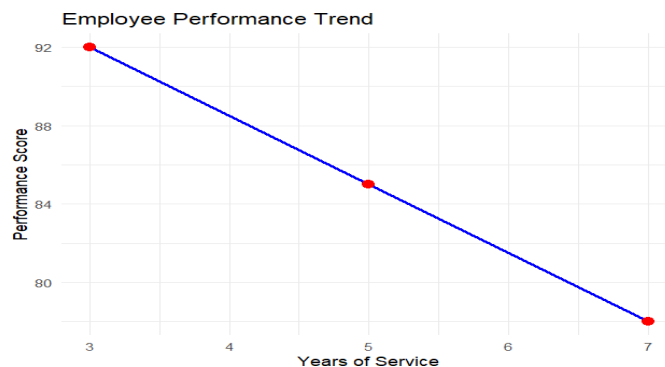
employee <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)

# Line Chart

ggplot(employee, aes(x = Years_of_Service, y = Performance_Score, group = 1)) +
  geom_line(color = "blue", linewidth = 1) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Employee Performance Trend",
    x = "Years of Service",
    y = "Performance Score"
  ) +
  theme_minimal()

```

Output:



2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

Program:

```

# Load library
library(ggplot2)

# Create dataset
employee <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),

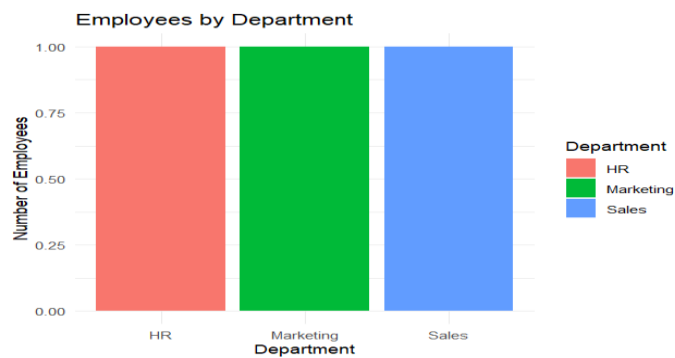
```

```

Years_of_Service = c(5, 3, 7),
Performance_Score = c(85, 92, 78)
)
# Bar Chart
ggplot(employee,
  aes(x = Department,
    fill = Department)) +
  geom_bar() +
  labs(
    title = "Employees by Department",
    x = "Department",
    y = "Number of Employees"
  ) +
  theme_minimal()

```

Output:



3. Build a table to display the performance data for each employee. Label the table elements.

Program:

```

# Create dataset
employee <- data.frame(
  Employee_ID = c(1, 2, 3),
  Department = c("Sales", "HR", "Marketing"),
  Years_of_Service = c(5, 3, 7),
  Performance_Score = c(85, 92, 78)
)
# Display table
print(employee)

```


Output:

Employee_ID Department Years_of_Service Performance_Score

1	1	Sales	5	85
2	2	HR	3	92
3	3	Marketing	7	78

set 9: Online_Learning_Activity.csv

1. Import the dataset in R. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by Course. Add title and labels. Interpret student performance.

Program:

```
# Load library
library(ggplot2)

# Create dataset
learning <- data.frame(
  Student_ID = c("L01","L02","L03","L04","L05","L06"),
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Age = c(20,22,19,21,23,20),
  Course = c("R","R","SQL","R","R","SQL"),
  Study_Time = c(3.5,4.2,2.0,5.0,2.5,4.0),
  Videos_Watched = c(12,15,8,18,9,14),
  Quiz_Score = c(78,85,65,92,70,88),
  Login_Date = c("2025-01-05","2025-01-05","2025-02-08",
    "2025-02-08","2025-03-12","2025-03-12")
)

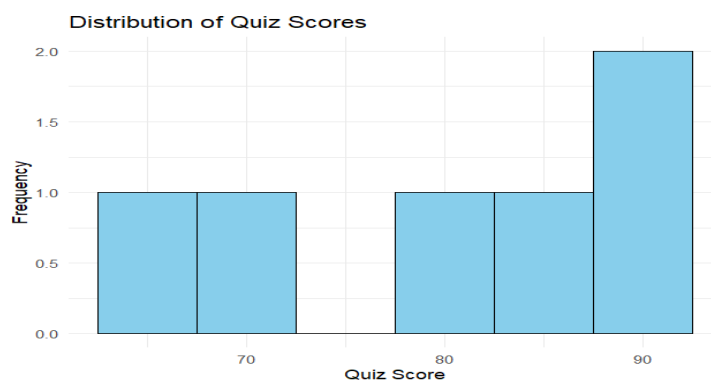
# Histogram
ggplot(learning, aes(x = Quiz_Score)) +
  geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +
  labs(
    title = "Distribution of Quiz Scores",
    x = "Quiz Score",
    y = "Frequency"
  ) +
```

```

theme_minimal()
# Boxplot
ggplot(learning,
       aes(x = Course,
           y = Quiz_Score,
           fill = Course)) +
geom_boxplot() +
labs(
  title = "Quiz Scores by Course",
  x = "Course",
  y = "Quiz Score"
) +
theme_minimal()

```

Output:



2. Create a scatter plot using R programming between Study_Time and Quiz_Score. Represent Videos_Watched using bubble size. Apply transparency. Explain the learning pattern.

Program:

```

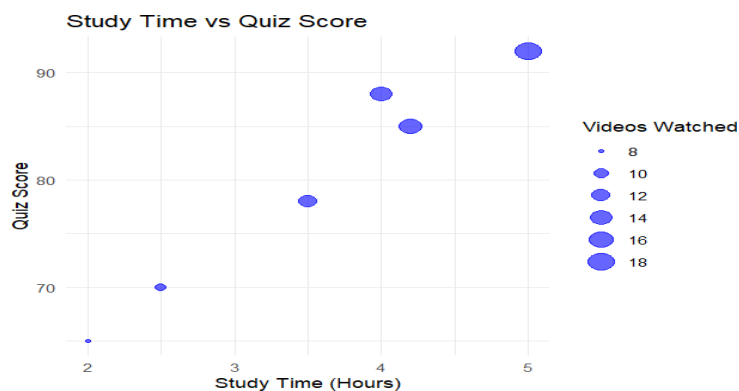
# Load library
library(ggplot2)

# Create dataset
learning <- data.frame(
  Student_ID = c("L01", "L02", "L03", "L04", "L05", "L06"),
  Study_Time = c(3.5, 4.2, 2.0, 5.0, 2.5, 4.0),
  Quiz_Score = c(78, 85, 65, 92, 70, 88),
  Videos_Watched = c(12, 15, 8, 18, 9, 14)
)

```

```
)
# Bubble Scatter Plot
ggplot(learning,
  aes(x = Study_Time,
    y = Quiz_Score,
    size = Videos_Watched)) +
geom_point(alpha = 0.6, color = "blue") +
labs(
  title = "Study Time vs Quiz Score",
  x = "Study Time (Hours)",
  y = "Quiz Score",
  size = "Videos Watched"
) +
theme_minimal()
```

Output:



3. Using R programming, convert Login_Date into Date format. Calculate average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.

Program:

```
# Load libraries
library(ggplot2)
library(dplyr)

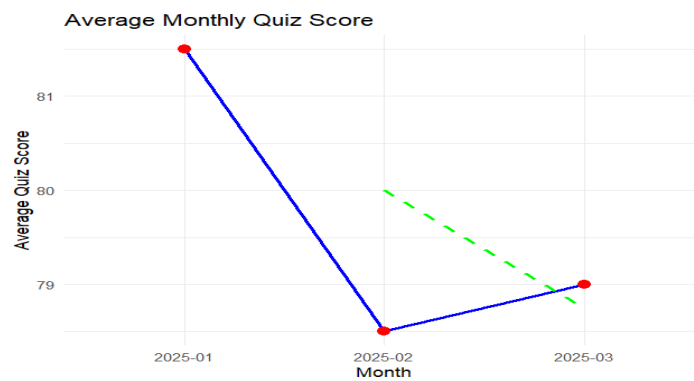
# Create dataset
learning <- data.frame(
  Student_ID = c("L01", "L02", "L03", "L04", "L05", "L06"),
  Quiz_Score = c(78, 85, 65, 92, 70, 88),
```

```

Login_Date = c("2025-01-05","2025-01-05","2025-02-08",
               "2025-02-08","2025-03-12","2025-03-12")
)
# Convert to Date
learning$Login_Date <- as.Date(learning$Login_Date)
# Extract Month
learning$Month <- format(learning$Login_Date, "%Y-%m")
# Average Quiz Score per Month
monthly_avg <- learning %>%
  group_by(Month) %>%
  summarise(Average_Score = mean(Quiz_Score))
# Moving Average
monthly_avg$Moving_Average <- stats::filter(
monthly_avg$Average_Score,
  rep(1/2, 2),
  sides = 1
)
# Line Chart
ggplot(monthly_avg, aes(x = Month, y = Average_Score, group = 1)) +
  geom_line(color = "blue", linewidth = 1) +
  geom_point(size = 3, color = "red") +
  geom_line(aes(y = Moving_Average),
            color = "green",
            linewidth = 1,
            linetype = "dashed") +
  labs(
    title = "Average Monthly Quiz Score",
    x = "Month",
    y = "Average Quiz Score"
  ) +
  theme_minimal()

```

Output:



Set 10: Survey Responses Analysis

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

Program:

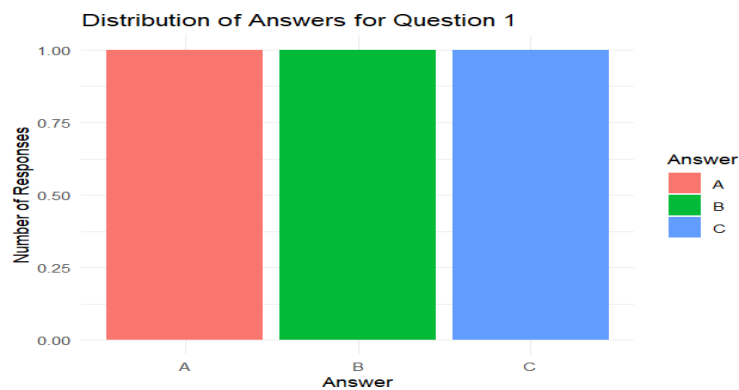
```
# Load library
library(ggplot2)

# Create dataset
survey <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question1 = c("A", "B", "C"),
  Question2 = c("B", "A", "A"),
  Question3 = c("C", "D", "B")
)

# Count responses for Question 1
q1 <- as.data.frame(table(survey$Question1))
colnames(q1) <- c("Answer", "Count")

# Grouped Bar Chart
ggplot(q1, aes(x = Answer, y = Count, fill = Answer)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Distribution of Answers for Question 1",
    x = "Answer",
    y = "Number of Responses"
  ) +
  theme_minimal()
```

Output:



2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

Program:

```
# Load libraries
library(ggplot2)
library(tidyr)

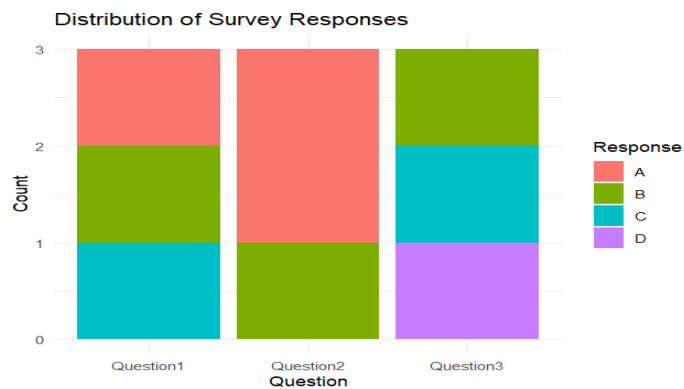
# Create dataset
survey <- data.frame(
  Survey_ID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)

# Convert to long format
survey_long <- pivot_longer(
  survey,
  cols = c(Question1, Question2, Question3),
  names_to = "Question",
  values_to = "Response"
)

# Stacked Bar Chart
ggplot(survey_long,
  aes(x = Question,
    fill = Response)) +
```

```
geom_bar() +
labs(
  title = "Distribution of Survey Responses",
  x = "Question",
  y = "Count"
) +
theme_minimal()
```

Output:



3. Build a table to show the survey response data for each survey. Label the table elements.

Program:

```
# Create dataset
survey <- data.frame(
  Survey_ID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)
# Display table
print(survey)
```

Output:

	Survey_ID	Question1	Question2	Question3
1	1	A	B	C
2	2	B	A	D
3	3	C	A	B

Set 11: Product Category Analysis

1. Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.

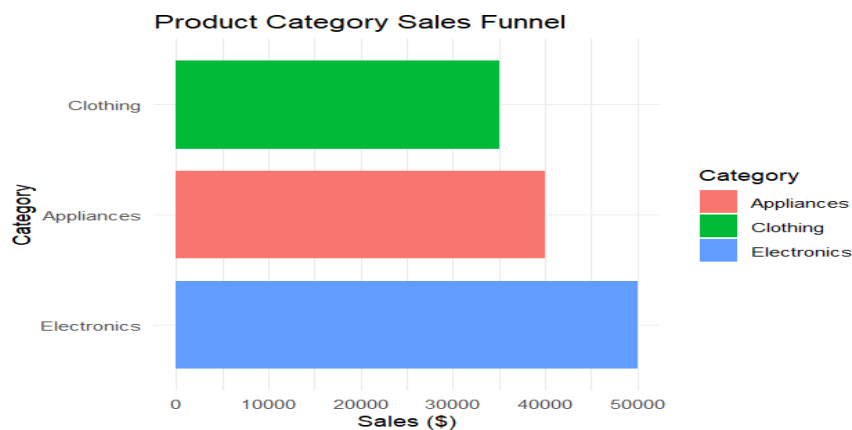
Program:

```
# Load library
library(ggplot2)

# Create dataset
category <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)

# Funnel Chart
ggplot(category,
  aes(x = reorder(Category, -Sales),
    y = Sales,
    fill = Category)) +
  geom_bar(stat = "identity", width = 0.8) +
  coord_flip() +
  labs(
    title = "Product Category Sales Funnel",
    x = "Category",
    y = "Sales ($)"
  ) +
  theme_minimal()
```

Output:



2. Build a table to display the sales data for each product category. Label the table elements.

Program:

```
# Create dataset
category <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)
# Display table
print(category)
```

Output:

```
Category Sales
1 Electronics 50000
2 Clothing 35000
3 Appliances 40000
```

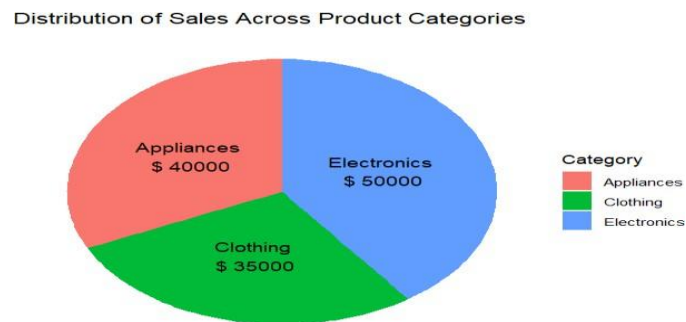
3. Create a pie chart to represent the distribution of sales across product categories. Include labels.

Program:

```
# Load library
library(ggplot2)
# Create dataset
category <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)
# Pie Chart
ggplot(category, aes(x = "", y = Sales, fill = Category)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar(theta = "y") +
  geom_text(
    aes(label = paste(Category, "\n$", Sales)),
    position = position_stack(vjust = 0.5),
    size = 4
  ) +
```

```
labs(
  title = "Distribution of Sales Across Product Categories"
) +
theme_void()
```

Output:



Set 12: Dataset: Website Traffic

1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

Program:

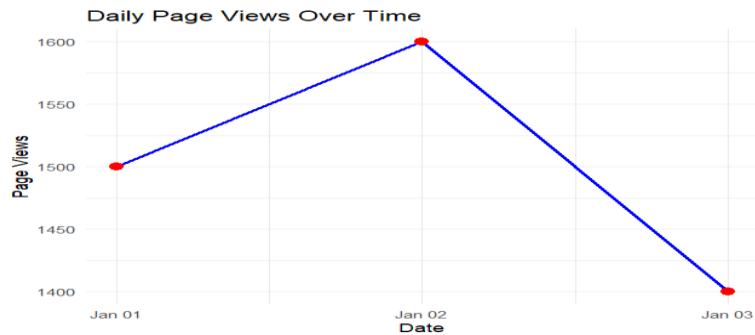
```
# Load library
library(ggplot2)

# Create dataset
traffic <- data.frame(
  Date = as.Date(c("2023-01-01",
                   "2023-01-02",
                   "2023-01-03")),
  Page_Views = c(1500,1600,1400),
  Click_Through_Rate = c(2.3,2.7,2.0)
)

# Line Chart
ggplot(traffic,
  aes(x = Date,
      y = Page_Views,
      group = 1)) +
  geom_line(color = "blue", linewidth = 1) +
  geom_point(color = "red", size = 3) +
```

```
labs(
  title = "Daily Page Views Over Time",
  x = "Date",
  y = "Page Views"
) +
theme_minimal()
```

Output:



2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

Program:

```
# Load library
library(ggplot2)

# Create dataset
traffic <- data.frame(
  Date = as.Date(c("2023-01-01",
    "2023-01-02",
    "2023-01-03")),
  Click_Through_Rate = c(2.3,2.7,2.0)
)

# Sort data
top_days <- traffic[order(-traffic$Click_Through_Rate), ]

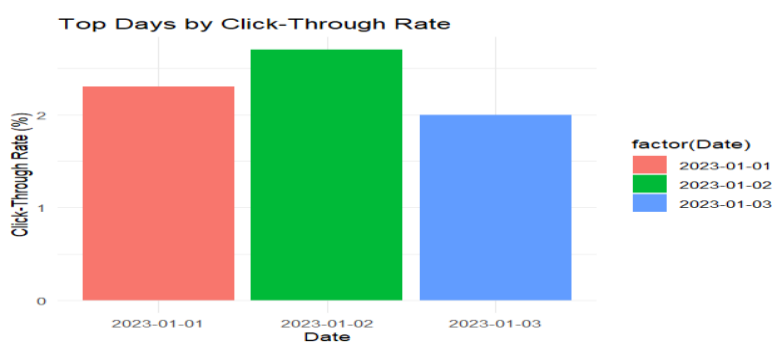
# Bar Chart
ggplot(top_days,
  aes(x = factor(Date),
    y = Click_Through_Rate,
```

```

fill = factor(Date))) +
geom_bar(stat = "identity") +
labs(
  title = "Top Days by Click-Through Rate",
  x = "Date",
  y = "Click-Through Rate (%)"
) +
theme_minimal()

```

Output:



3. Build a table to show the website traffic data for each day. Label the table elements.

Program:

```

# Create dataset

traffic <- data.frame(
  Date = c("2023-01-01",
           "2023-01-02",
           "2023-01-03"),
  Page_Views = c(1500,1600,1400),
  Click_Through_Rate = c("2.3%", "2.7%", "2.0%")
)

# Display table
print(traffic)

```

Output:

	Date	Click_Through_Rate
1	2023-01-01	2.3
2	2023-01-02	2.7
3	2023-01-03	2.0